Multimedia Tools and Applications (2022) 81:43837–43849
https://doi.org/10.1007/s11042-022-13215-1

Otsu's thresholding technique for MRI image brain tumor segmentation

Myat Thet Nyo[1] · F. Mebarek-Oudina[2] · Su Su Hlaing[3] · Nadeem A. Khan[4]

Received: 2 March 2021 / Revised: 17 February 2022 / Accepted: 11 May 2022 /
Published online: 30 May 2022
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2022

Abstract
MRI image segmentation is very challenging area in medical image processing. It is implemented with the low contract of MRI scan. In terms of certain input features or expert information, the major objective of medical image segmentation is to isolate and describe anatomical constitutions. In MRI image segmentation, brain tumor segmentation is more difficult because of its complex structure. The Otsu's thresholding method is well-known method in image segmentation. In this paper, choosing the classes or bins of Otsu's thresholding are analyzed on MRI image brain tumor segmentation. As a preprocessing, the 2D MRI images are convert the grayscale image and resized to the same size. And then, median filter is utilized to eliminate the noise from MRI image. In MRI image segmentation, the varieties of classes or bins of Otsu's thresholding are utilized to segment the brain tumor from MRI images. Then, the morphological operation is used to achieve the accurate tumor regions. All of the experiments are tested on 2015 BRATS dataset. As segmentation quality validation metric, Jaccard similarity index, true positive rate (Sensitivity), true negative rate (Specificity) and accuracy are used to validate the segmented results and their ground truth. According to the results, level 4 or class 4 got 68.7955% in true positive and 95.5593% in accuracy. Class 4 is the best or suitable for MRI image segmentation according to experiments.

Keywords MRI · Median filter · Otsu's thresholding · BRATS

✉ F. Mebarek-Oudina
f.mebarek_oudina@univ–skikda.dz

[1] Faculty of Computer Science, University of Computer Studies, Pakokku, Myanmar

[2] Department of Physics, Faculty of Sciences, University of 20 août 1955-Skikda, Skikda, Algeria

[3] Faculty of Information Science, Myanmar Institute of Information Technology, Mandalay, Myanmar

[4] Department of Civil Engineering, Jamia Millia Islamia, -110025, New Delhi, India

1 Introduction

Through encouraging the delineation of anatomical structures or automating, medical imaging applications play an indispensable position. Due to the different features of the images, medical image segmentation is a difficult task, which adds to the difficulty of segmentation. Magnetic resonance imaging is commonly used for medicinal diagnosis, disease development and monitoring in hospitals and clinics without revealing the body to ionizing radiation. It is used for the diagnosis of problems such as cancers, bleeding, fractures, vascular disorders or infections. Tumors are a group of abnormal cells that form masses or growths.

Image segmentation is the method of dividing a digital image into numerous segments. Threshold segmentation is widely utilized in several regions due to its usefulness and simplicity. Thresholding is a method mostly applied to the segmentation of images. Its fundamental objective is to divide a given image's pixels into two classes: those related to an object and those related to the context. Tumor segmentation is mainly intended to detect and estimate the size and shape of the tumor. The precision of segmentation can help to define the precision of tumor classification and further evaluate the types of tumors.

The goal of this research is to create a new threshold approach that is more efficient and straight forward to use than existing threshold techniques. The proposed threshold technique is compared to one of the most widely used threshold methods, the OTSU. In this analysis paper, analysis classes or bins of Otsu's thresholding are described and presented. We discovered that Otsu's thresholding technique provides better accuracy and takes less time than the other approaches.

This manuscript is organized as follows. In Section 2, Related work is illustrated. In Section 3, material and methods is explained. In Section 4, experimental outcomes are presented and Section 5 is conclusion of the paper.

2 Related work

Bindu [4] presented the paper with the title "An improved medical image segmentation algorithm using Otsu method". This work describes a way of segmenting medical images utilizing an optimized Otsu approach based on an enhanced thresholding algorithm. The experimental results of this algorithm demonstrate that the current optimized approach significantly reduces running time and improves the separability factor in the segmentation of medical images while maintaining the accuracy of final image segmentation. For medical image segmentation, the improved Otsu thresholding method has been proposed in this manuscript. This technique yields better results than the other methods of thresholding and produces sufficient binary images that can be used in more processing phases.

Sasilatha et al. [20] introduced about an efficient strategy through brain tumor analysis utilizing image processing techniques. The overall steps of this paper are tumor segmentation, morphological operation, feature extraction and classification. In this paper, based on symmetrical analysis, the tumor affected region is determined with several data sets with distinct tumor size, intensity and position. They demonstrated that their algorithm is able to identify and segment the brain tumor automatically. In the segmentation methods, Otsu's thresholding method is utilized to segment the region of interest. And then, erosion and dilation are used as morphological operation. In feature extraction, texture feature, align-based feature, edge-based feature and context aware feature are extracted. KNN classifier is used in image classification.

Srikanth and Bikshalu [22] presented the paper with the title is "Segmentation of low contrast satellite and medical images based on level set function with harmony search optimization algorithm". In this paper, medical and satellite images considered for image segmentation to remove features are low contrast images. This paper presents a novel method of image segmentation by Level Set Function along through Harmony Search Algorithm (LSFBHEHS)-based Bi-Histogram Equalization. In order to find the optimized threshold level to divide the histogram into two sections, the multilevel thresholding technique utilized by Otsu to find threshold level is used to find optimized threshold level Harmony Search Algorithm (HSA) as an objective feature to maximize inert class variance. The qualitative calculation used such as Measure of Enhancement (EME), Dice similarity index and time needed for the evaluation of the proposed method and SBGFRLS, numerous low contrast medical images and satellite are considered for experimentation, outcomes explained that the proposed technique is more effective for low inhomogeneous and contrast images. According to the results, the proposed system achieved the satisfied results.

Bindu and Prasad [5] presented the paper with the title is "An efficient medical image segmentation using conventional OTSU method". In this manuscript, a hybrid algorithm based on a self-adaptive thresholding technique is proposedto optimize the threshold of the Otsu's method. Time-consuming computation is the conventional Otsu approach for medical image segmentation and has become an obstacle in real-time application systems. The TSMO approach was also compared in the same manner. This paper describes a way of segmenting medical images using an optimized Otsu approach based on an enhanced thresholding algorithm. The experimental outcomes show that the Otsu threshold technique proposed can be easily achieved with a better image segmentation result. The findings also indicate that one of the suggested methods is effective relative to the current system. But the proposed method is more time complexity among the traditional Otsu's thresholding.

Gaikwad et al. [8] suggested a paper with the title is "Medical image segmentation for brain tumor detection". The new strategy for image segmentation is presented in this paper. Canny's image segmentation algorithm is combined with Otsu's efficiency and result enhancement algorithm. For the clear appearance of the tumor or abnormal image part, the Canny operator and image thinning algorithm are used. The system first filters a photo of the brain with the tumor. Authors then apply the Otsu algorithm to separate the part of the tumor from the initial image. Retrieve the edge map image at the same time by using the Canny operator on the original image. The thinning techniques are then used by the system to obtain the thin edge of the image. Use the area rising method to paint the tumor portion of the brain. These two images, after all, are combined with the original image containing the tumor as the final output. Both abnormal and regular MRI images have been the subject of many studies. Authors found this system to be very flexible; it was simple. This mechanism is also used to define the nature of the bone and to classify blood cells. This system is found to be very compatible with a lot of medical image segmentation during these experiments. But the interest in the discovery of brain tumors was, and we are satisfactory achievements.

Min and War [16] presented the comparative paper on brain tumor segmentation methods from MRI image. Otsu thresholding, region growing, particle swarm optimization and interactive graph cut segmentation techniques are implemented in this comparative paper and they are experienced in the BRATS dataset on 72 flair images.Ture positive rate (TPR), True negative rate (TNR), Accuracy and Jaccard Similarity index are used as the assessment techniques. The runtime duration of each method is also

described in this paper. According to the test results, without morphological activity, area rising segmentation is more accurate than other methods. The fact indicates that without morphological action, this technique can segment well. Otsu's thresholding method achieved better results than other approaches of morphological operation. The segmentation outcomes of particle swarm optimization do not vary greatly with region growth in MRI image segmentation. When comparing another technique in MRI image segmentation, interactive graph cut segmentation requires more time and less precision. Finally, threshold-based methods and graph-based segmentation methods are better than other region-based methods.

Sheeba and Manikandan [21] suggested the paper about image segmentation utilizing bi-level thresholding. For image segmentation, four image thresholding methods are carried out in this work, namely Between Class Variance (Otsu's), Total Class Variance (Hou's), Posterior Maximum Entropy (kapur's) and Minimum Error Thresholding.For the above approaches, the Pixel-based output measurements are measured for False Positive Rate, Misclassified Pixels, Average Absolute Error and False Negative Rate. The process of Posterior maximum entropy provides better results than the other techniques. For threshold selection, two popular statistical methods, i.e., Otsu's and Hou's techniques, choose class variance criteria. Both techniques take account of the variance sum of the object and background classes only. Consequently, these approaches do not have adequate results.

Kazi et al. [13] presented the paper about MRI brain image segmentation utilizing adaptive thresholding and k-means algorithm. A new technique based on adaptive thresholding and K-means clustering algorithm has been introduced in this paper, which is utilized to get Gray Matter (GM), cerebrospinal fluid (CSF), White Matter (WM) and others. An image thresholding method is adopted in order to segment but a fixed threshold is not appropriate for segmentation, if the context is rough, adaptive thresholding technique is therefore more appropriate for segmentation and segmentation is often used by the K-means clustering algorithm. Hear adaptive thresholding and K-means clustering have performed best in generating various input MRI image clusters. Many state-of-the-art methods are over looked in Table 1.

3 Methods section - background theory

This section describes about the theoretical background of methods [1]. They are gray scale converting, median filter, Otsu's thresholding and morphological operation. Based on literature review, classification of MRI brain image segmentation methods is depicted in Fig. 1. Various MRI brain image segmentation methods available in literature, we discovered that Otsu's thresholding technique provides better accuracy and takes less time than the other approaches.

A. **Grayscale Converting**

The system accepts the RGB MRI image from BRATS dataset with JPEG format. In noise removal and segmentation process are worked on grayscale image. Therefore, the system needs to convert RGB to grayscale image. In grayscale converting, NTSC color model is used in this system. The formulation of NTSC color model is described in Eqs. (1) and (2).

Table 1 Key findings in many state-of-the-art

References	Key Findings
Gao et al. [9]	Each year, around 12.7 million individuals are diagnosed through cancer, with 7.6 million dying from it. After stomach cancer, uterine cancer, breast cancer, and esophageal cancer, brain cancer ranks fifth in terms of incidence and death.
Lois et al. [15]	The area, tissue type, non-cancerous or cancerous (benign or malignant) cells, site of origin (primary or secondary), and other essential variables are used to classify brain tumors.
Ozols et al. [18]	Medical imaging techniques such as positron emission tomography, (PET), magnetic resonance imaging (MRI), and compute d tomography (CT) are utilized to diagnose brain cancers.
Prince and Links [19]	PET injects radioactive medicines into the human body, and the drugs circulate through the bloodstream to all cells, tissues, and organs throughout the body.
Xuan and Liao [27]	There are no tools or medications injected into the human body during the MRI procedure. The human body is not exposed to radiation, and the entire procedure is completely safe. Furthermore, because MRI has excellent resolution and accurate placement of soft tissues, as well as sensitivity to illness characteristics, it is particularly well suited to the detection of brain diseases.
Mishra et al. [17]	In the medical area, image processing is one of the most widely utilized and effective techniques. Medical image processing aims to build solutions that use computerized technologies to tackle medical diagnosis challenges.
Taheri et al. [24]	TLS is a threshold-based method for 3D tumor segmentation in brain MRI data.
Ali et al. [2]	To calculate the tumor's area, researchers used the Enhanced Thresholding Algorithm.
Vijayarangan [26]	Using segmentation and histogram thresholding, a technique for detecting brain tumors in MRI images has been developed.
Anandgaonkar and Sable [3]	Implemented a brain tumor detection and identification algorithm based on Fuzzy C-Means clustering to adaptive thresholding to extract the tumor and partition the MRI into a number of clusters, then choosing the type of tumor based on its area.
Sujan et al. [23]	Developed a method for detecting brain tumors from MRI images using thresholding and morphological image processing approaches. They converted the MRI image to grayscale, removed noise with various filtering approaches, then converted the new image to binary using Otsu's threshold method for segmentation, and then did morphological operations to find the tumor that included the brightest area of the image.
Kaur et al. [12]	The authors have suggested a methodology for uncovering information in a database by exposing hidden patterns that can aid in making reliable decisions. Using a variety of illness-specific input features, this paper examined prediction algorithms for diseases such as "heart disease", "breast cancer", "liver disorders", "diabetes", "thyroid", "dermatology", "spect heart", and "surgical data".
Dargan et al. [7]	Business, science, and government are the most widely used domains of deep learning, which also include "adaptive testing", "natural language processing", "computer vision", "cancer detection", "object detection", "face recognition", "smart city", "handwriting recognition", "speech recognition", "stock market analysis", "biological image classification" and many more. The ideas of deep learning, as well as their basic and advanced structures, methodologies, motivational factors, traits, and limitations, are discussed in this work.
Dargan and Kumar [6]	This article describes a comprehensive and in-depth survey that concisely and systematically summarizes the literature work on "unimodal and multimodal biometric systems and analyses feature extraction techniques", "datasets", "classifiers", "efficiency", "results", and "reliability of the system" from high and multi-dimensional perspectives.

Table 1 (continued)

References	Key Findings
Ghosh et al. [10]	In this work, authors examine ten commonly used filter ranking techniques in depth. Authors assessed the accuracies of the approaches on ten microarray datasets (both binary and multi-class) using three well-known classifiers: "Multi-layer Perceptron (MLP)", "Support Vector Machine (SVM)", and "K-Nearest Neighbor (K-NN) (KNN)".
Kumar et al. [14]	This paper provides a method for detecting faces in still photos with occlusion and non-uniform light. The authors describe a face detection method that combines the YCbCr, HSV, and L a b color models. Accuracy, Detection Rate, False Detection Rate, and Precision were all improved with the proposed technique.
Gupta et al. [11]	The authors describe a feature-based technique for 2D face photographs in this research. SURF (sped up robust features) and SIFT (scale-invariant feature transform) are used to extract features. Yale2B, Face 94, M2VTS, ORL, and FERET are five public datasets that were used in the experiment.

$$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ 0.596 & -0.274 & -0.322 \\ 0.211 & -0.523 & 0.312 \end{bmatrix} \tag{1}$$

The sum of the first row is equal to 1, while the sum of the remaining two rows is 0. In this system, the only Luminance component is used as the enhanced image information.

$$\widehat{I}G = Y = 0.299\,R + 0.587\,G + 0.114\,B \tag{2}$$

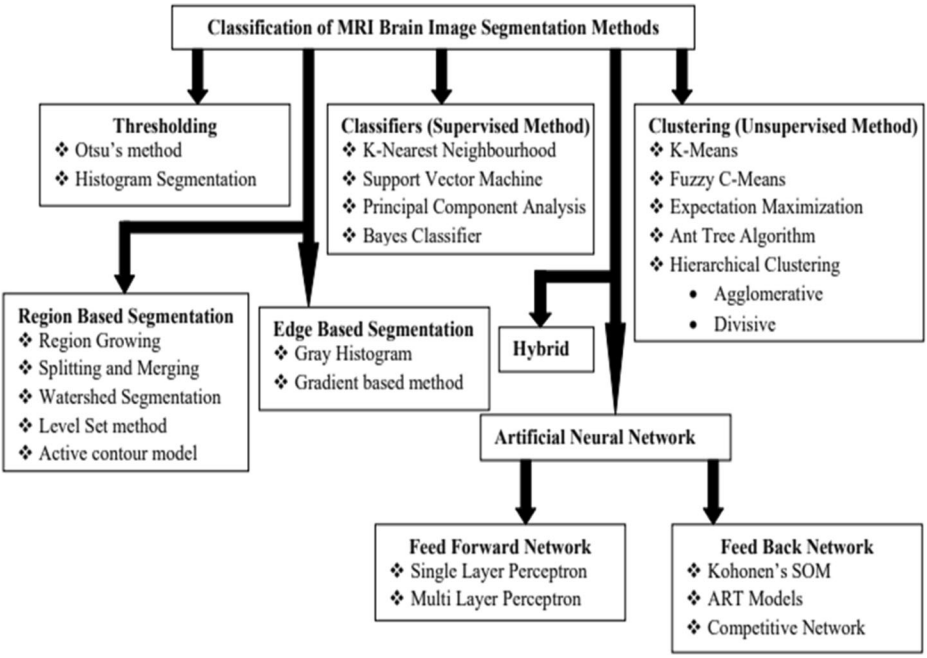


Fig. 1 Classification of MRI brain image segmentation methods

where, $\widehat{I}G$ is the result of gray image, G is green color component, R denotes red color component and B presents blue color component.

B. Median Filter

Median filtering is a nonlinear technique of removing noise from images of the MRI brain. The removal of salt and pepper noise is especially efficient. Median filter operates by pixel shifting the image pixel, changing each value with the adjacent pixel median value. The pixel is calculated by first separating all the pixel values from the neighborhood pattern into mathematical order and then changing the pixel that is regarded as the average pixel value. Median filter can extract noise better without decreasing picture sharpness. The describing the formulation of Median filter as follow:

$$f(x,y) = median\ \{g(s,t)\} \qquad where\ (s,t) \in S_xy \tag{3}$$

where, $S\ _\ xy$ is the set of coordinates in a rectangular sub-MR image window which has center at $(x,\ y)$. $f(x,\ y)$ is the restored image and $g(s,\ t)$ is corrupted and calculated area under the $S\ _\ xy$.

C. Otsu's Thresholding

Assuming the segment has relatively the same gray level value and the threshold T can be selected to minimize the dispersion of the gray level in the segment or minimize the adjustment between the object and the back ground k. This method makes then most of the variance between classes and is based on calculations accomplished on the image histogram.

Certainly, the algorithm can be implemented in O(LN) time. It is not difficult to make it run in O(N + L^2) time as shown in Table 2. Make a histogram of the pixel intensities (count of the number of pixels of each intensity). Then, rather than taking O(N) time to compute each variance, you can do so in O(L) time. Assume we've calculated the sum of values below the threshold and the sum of squared values below the threshold, as well as the sum of values above the threshold. To compute the standard deviations, this is all that is required. Then, by increasing the threshold by one, we may update those four sums in O(1) time. As a result, building the histogram takes O(N) time, then computing the score for the first threshold takes O(L) time, then O(1) time for each successive threshold takes O(1) time, and there are L

Table 2 Algorithm of Otsu's Thresholding

Algorithm of Otsu's Thresholding	
Step.1	Compute the standardized histogram of the input image. The components of the histogram are denoted via Pi=ni / MN, where i=0, 1, 2, L-1 and MN=n0+n1+n2+…+nL-1
Step.2	Calculate the cumulative sums P1(k)= # $ %&, for k=0,1,2,…, L-1
Step.3	Calculate the aggregate means,
Step.4	Calculate the global intensity mean, mG
Step.5	Calculate the between-class variance,
Step.6	Get the threshold Otsu k * as the maximum value of k. If the maximum is not unique, k * is obtained through averaging the values of k via to the diverse maximum values found.

candidate thresholds, totaling O(N + L) time. In fact, the difference between O(N + L^2) and O(N + L) may be insignificant, especially for 8-bit pictures, when L < =256 is used.

D. Morphological Operation

In this section, the system used the opening and closing operation as the morphological operation. The opening operation erodes an image and then dilates the eroded image, with both operations utilizing the similar structuring element. For removing small objects from an image while maintaining the shape and size of larger objects in the image, morphological opening is helpful. The closing operation dilates an image and then erodes the dilated image, with both operations utilizing the same structuring element. Morphological closing is useful for filling a picture with small holes while maintaining the shape and preserving the form. The formulations of opening and closing are described in Eqs. (4) and (5) respectively.

$$A \circ B = (A \Theta B) \oplus B \tag{4}$$

$$A \bullet B = (A \oplus B) \Theta B \tag{5}$$

where A is input image and B is structuring element and denote Θ erosion and ⊕ dilation, respectively.

4 System implementation

This system is analysis system on thresholding-based MRI image segmentation. This system is implemented with four steps. They are preprocessing, noise removing, segmentation and morphological operation. The architecture of the system is described in Fig. 2.

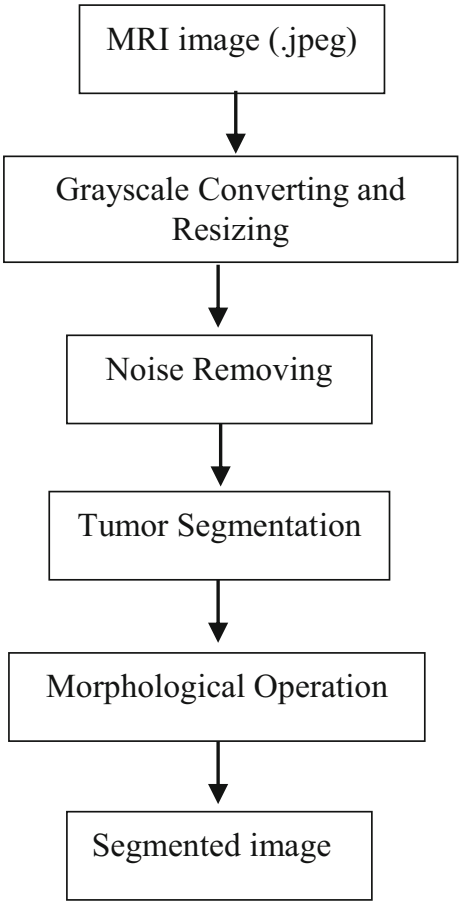
The system accepts the RGB image and convert RGB to grayscale image. And then, the system resizes the grayscale image to get 240 × 240 dimension. Median filter is used in noise removing step. Otsu's thresholding is used in tumor segmentation and morphological operation is used to achieve the accurate tumor shape and size. Finally, the system generates the tumor segmented image.

5 Experimental results and discussion

In this part, the segmented results are described. All the experimental results are tested on HGG, LGG data folder of 2015 BRATS dataset. In this data folder, 110 Flair images include. In this data folder, the Flair sequence describe the whole tumor area. Therefore, the system is tested and evaluated on Flair sequence. The evaluation metrics of image segmentation are described as the following equations.

All of the photographs in the database were put to the test in this investigation. The following attributes are utilized in the computations to evaluate the system's accuracy, sensitivity, and specificity (as shown by Table 3).

Fig. 2 Architecture of the System



Sensitivity is a metric for determining whether or not a person has a tumor. Sensitivity is determined by

$$Sensitivity\ (\%) = \frac{TruePositive}{TruePositive + FalseNegative}*100 \tag{6}$$

Specificity is a metric for determining whether or not a person has a tumor. The following factors contribute to the specificity:

Table 3 Attributes for system performance

Attributes	Description
True Positive ($T_{rue}P_{ositive}$)	Existing tumor that was accurately identified
True Negative (T_{rue}Negative)	Tumor that did not exist and was not detected
False Positive ($F_{alse}P_{ositive}$)	Detection of a previously undetected tumor
False Negative ($F_{alse}N_{egative}$)	Existing tumor that hasn't been detected

$$Specificity\ (\%) = \frac{TrueNegative}{TrueNegative + FalsePositive} * 100 \quad (7)$$

The accuracy of a classification is the yardstick by which it is judged. The precision is determined by:

$$Accuracy\ (\%) = \frac{TruePositive + TrueNegative}{TruePositive + TrueNegative + FlasePositive + FalseNegative} * 100 \quad (8)$$

$$Jaccard(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (9)$$

In analysis on Otsu's thresholding, the 2, 3 and 4 classes or bins are tested and presented in this section. Table 2 depicts the average outcomes of 110 flair images. Figure 3 describes about the average results of 110 flair image with bar chart.

According to the results of Table 4, the segmentation results of class 4 of Otsu's thresholding are the best among the rest classes segmentation results. It's results of true positive rate true negative rate, accuracy and Jaccard similarity index is more than other classes. Class 2 got the smallest indication index in this comparison. Due to the experimental results, Class 4 is the best class of Otsu's thresholding method for MRI image segmentation. Table 6 describe run time execution of the classes and Fig. 4 shows the results of run time execution with line chart. The comparison's outcome is shown in Table 5.

In the comparison of run time execution, the class 2 spends the shortest time in run time execution. Class 3 is the second shortest time in run time execution and class 4 spends the maximum time in run time execution. Although class 4 spends run time execution, it gets the best results in tumor segmentation (Table 6).

6 Conclusion

In this paper, the classes or bins of Otsu's thresholding method are analyzed and evaluated from MRI image segmentation. The Otsu's thresholding technique is one of very efficient

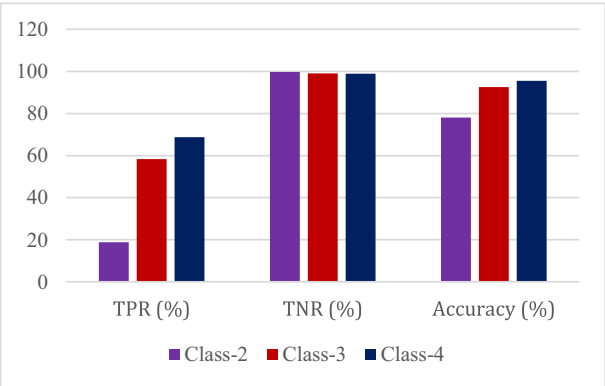


Fig. 3 Average Segmentation Result of Flair

Table 4 Averages Segmentation Results of Flair

Classes	TPR (%)	TNR (%)	Accuracy (%)	Jaccard
Class-2	18.8105	99.8060	78.0420	0.1870
Class-3	58.2573	99.1271	92.5531	0.5032
Class-4	68.7955	98.9013	95.5593	0.5650

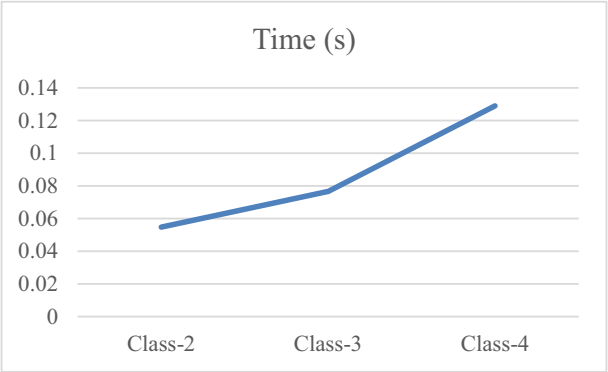


Fig. 4 Average Run Time Execution

Table 5 The proposed system's comparison

References	Methodology	Accuracy
Sujan et al. [23]	OTSU's threshold	84.72%
Tanuja and Shewale [25]	Region growing	80%
Proposed	Proposed threshold	95.5593%

techniques to threshold the gray images. However, its computation would become more difficult. The computation strongly depends on the level of class of bin. According to the results, level 4 or class 4 got 68.7955% in true positive and 95.5593% in accuracy. Class 4 is the best or suitable for MRI image segmentation according to experiments. In run time duration, run time of class 4 is not so different from class 2 and class 3. Therefore, class 4 is the suitable level of Otsu's thresholding for MRI image segmentation. There one flaw in our system that need to be improved for future development. The method proposed is not a fully automated procedure. These technologies could be combined in one processing environment using neuro-fuzzy or support vector machine. In Future, we aim to obtain the safety boundary of the tumour and plotting its margin for operation planning. This will help doctors to make

Table 6 Averages Run Time Execution

Classes	Time (s)
Class-2	0.054628076
Class-3	0.076594666
Class-4	0.128929684

surgery based on automatic border calculation. Also, keep the patient safe from subsequent biopsy procedures. We intend to present a full method that includes EEG brain mapping to detect problems with greater precision.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

1. Ahmad S, Mebarek-Oudina F, Mehfuz S, Beg J (2022) RSM analysis based cloud access security broker: a systematic literature review. Clust Comput. https://doi.org/10.1007/s10586-022-03598-z
2. Ali AH, Khalaph KA, Nema IS (2014) IOSR J Res Method Educ (IOSR-JRME) 4 (1.-II): 58-62
3. Anandgaonkar G, Sable G (2014) Brain tumor detection and identification from T1 post contrast MR images using cluster-based segmentation. Int J Scid Res 3(4):814–817
4. Bindu CH (2009) An improved medical image segmentation algorithm using Otsu method. Int J Recent Trends Eng 2(3):88–90
5. Binduand CH, Prasad KS (2012) An efficient medical image segmentation using conventional OTSU method. Int J Adv Sci Tech 38:67–74
6. Dargan S, Kumar M (2020) A comprehensive survey on the biometric recognition systems based on physiological and behavioral modalities. Expert Syst Appl 143:113114
7. Dargan S, Kumar M, Ayyagari MR, Kumar G (2020) A survey of deep learning and its applications: a new paradigm to machine learning. Arch Computat Methods Eng 27:1071–1092
8. Gaikwad DP, Abhang P, Bedekar P (2011) Medical image segmentation for brain tumor detection. Proceedings of the International Conference & Workshop on Emerging Trends in Technology, Mumbai, pp 63–65
9. Gao Q, Wang XY, Qiu SJ, Yamato I, Sho M, Nakajima Y, Xu Y (2009) Over expression of PD-L1 significantly associates with tumor aggressiveness and post operative recurrence in human hepatocellular carcinoma. Clin Cancer Res 15(3):971–979
10. Ghosh KK, Begum S, Sardar A, Adhikary S, Ghosh M, Kumar M, Sarkar R (2021) Theoretical and empirical analysis of filter ranking methods: experimental study on benchmark DNA microarray data. Expert Syst Appl 169:114485
11. Gupta S, Thakur K, Kumar M (2021) 2D-human face recognition using SIFT and SURF descriptors of face's feature regions. Vis Comput 37:447–456
12. Kaur P, Kumar R, Kumar M (2019) A healthcare monitoring system using random forest and internet of things (IoT). Multimedia Tools Applications 78:19905–19916
13. Kazi IM, Chowhan SS, Kulkarni UV (2017) MRI brain image segmentation using adaptive thresholding and K-means algorithm. Int J Compr Appl 167(8):11–15
14. Kumar A, Kumar M, Kaur A (2021) Face detection in still images under occlusion and non-uniform illumination. Multimed Tools Appl 80:14565–14590
15. Louis DN, Ohgaki H, Wiestler OD, Cavenee WK, Burger PC, Jouvet A, Kleihues P (2007) The 2007 WHO classification of tumours of the central nervous system. Acta Neuropathol 114(2):97–109
16. Min A, War N (2019) Comparative Experiments of Brain Tumor Segmentation Methods in MRI image. Proceedings of the seventeenth international conference on computer applications (ICCA 2019), Yangon, Myanmar, Feb. 2019, pp. 287–293
17. Mishra A, Rai A, Yadav A (2014) Medical image processing: a challenging analysis. International Journal of Bio-Science and Bio-Technology 6(2):187–194
18. Ozols RF, Herbst RS, Colson YL, Gralow J, Bonner J, Curran WJ, Markman M (2006) Major research advances in cancer treatment, prevention, and screening report from the American Society of Clinical Oncology. J Clin Oncol 25(1):146–162
19. Prince JL, Links JM (2006) Medical imaging signals and systems. Upper Saddle River, New Jersey
20. Sasilatha T, Mahalakshmi R, Kumarand Dhanya PSM (2017) An efficient strategy over brain tumour analysis using image processing techniques. International Journal of Pharmacy & Technology 9(3):30688–30697
21. Sheeba A, Manikandan S (2014) Image segmentation using bi-level thresholding, proceedings of the 2014 international conference on electronics and communication systems (ICECS). IEEE, pp 1-5

22. Srikanth R, Bikshalu K (2019) Segmentation of low contrast satellite and medical images based on level set function with harmony search optimization algorithm. Int J Eng Adv Tech (IJEAT) 9:7306–7313
23. Sujan M, Alam N, Abdullah S, Jahirul M (2016) A segmentation based automated system for brain tumor detection. Int J Comp Appl 153(10):41–49
24. Taheri S, Ong SH, Chong VFH (2010) Level-set segmentation of brain tumors using a threshold-based speed function. Image Vis Comput 28(1):26–37
25. Tanuja PS, Patil SB (2016) Detection of brain tumor based on segmentation using region growing method. Int J Eng Innov & Res 5(2):173–176
26. Vijayarangan V (2014) Multiscale modeling for image analysis of brain tumor detection and segmentation using histogram thresholding. Int J Eng Comp Sci 3(8):7525–7534
27. Xuan X, Liao Q (2007) Statistical structure analysis in MRI brain tumor segmentation. Fourth International Conference on Image and Graphics. Sichuan, China, pp 421–426

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.